

Digital Twin Cloud Framework for Smart Water Distribution Optimization

Literature Survey and Research Gap Analysis

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No.	Paper Title (2023–2026)	Venue / Publisher	Method	Dataset / Case Study	Advantages	Limitations	Research Gap (Analytical Perspective)
1	Smart Water Grids and Digital Twin for the Management of System Efficiency in WDNs (2023)	MDPI (Water)	Digital Twin + Smart Water Grid	Real municipal water distribution network	Enables continuous real-time monitoring and operational efficiency gains	Lacks dynamic, AI-driven predictive capabilities for long-term forecasting	Most existing physical DT implementations focus strictly on monitoring; there is a need to integrate cloud-based predictive AI for dynamic leak forecasting and preventive maintenance.

2	Optimization of Water Distribution Systems Using Genetic Algorithms: A Review (2023)	Springer	Genetic Algorithms (GA) Optimization	Standard benchmark WDN datasets	Comprehensive evaluation of multi-objective optimization techniques	Heavy reliance on offline, batch-mode processing	Traditional GA optimizations are computationally heavy and static; they need to be adapted into scalable cloud microservices for real-time DT recalibration.
3	Machine Learning for Smart Water Distribution Systems: Exploring Applications and Challenges (2025)	Springer	Systematic ML Survey	High-frequency smart meter datasets	Deep breakdown of modern predictive ML applications in water management	Isolated ML models without operational DT integration	While ML applications are well-studied, there remains a gap in embedding these standalone models directly into cloud-managed IoT streaming architectures for real-time action.
4	Graph Neural Networks for Sensor Placement: Towards a Digital Twin of WDS (2024)	MDPI (Sensors)	Graph Neural Networks (GNN)	Simulated EPANET hydraulic networks	Captures complex spatial topology for optimal sensor layout	Focuses solely on static sensor positioning	Graph-based spatial modeling is currently restricted to design-time sensor placement; expanding GNNs to real-time transient detection in cloud DTs

							remains unaddressed.
5	Digital Twin Applications in the Water Sector: A Review (2025)	MDPI (Water)	Systematic Review	Global municipal water systems	Establishes a comprehensive taxonomy of DT maturity models	High proportion of conceptual designs with few field deployments	Literature is heavy on conceptual frameworks; there is a distinct lack of open, end-to-end cloud infrastructure blueprints built for production-scale water utilities.
6	Critical Drivers and Barriers of Digital Twin Adoption in Water Infrastructure (2025)	Wiley	ESG Framework + DEMATEL	Utility stakeholder surveys & literature	Identifies clear institutional, technical, and financial barriers	Lacks practical software architecture solutions	While socio-technical barriers are well-documented, actionable technical blueprints designed to lower cloud deployment overhead and operational complexity are missing.

7	Smart Water Management with Digital Twins and Multimodal Transformers (2025)	IEEE / Scopus	DT + Multimodal Transformers	Multi-sensor feeds (flow, pressure, thermal)	High accuracy in capturing non-linear spatiotemporal water demands	Extremely high computational overhead for localized processing	Complex deep learning models often stall at the prototype phase; distributed cloud inference pipelines are required to handle high-throughput multi-sensor data streams continuously.
8	A Systematic Literature Review of Digital Twin Applications for Sustainable Water (2026)	Springer	PRISM A Systematic Review	Global water utility case studies	Clear classification of architectural maturity across urban systems	Synthesis-driven without empirical framework validation	Current literature clearly categorizes DT theoretical levels, but practical open-source operational prototypes demonstrating real-time closing of the loop are scarce.
9	Digital Twins for Water Utilities: Architectures, Calibration Workflows and Benefits (2025)	IEEE / Scopus	Structural Architecture Review	Industrial water utility implementations	Clearly maps hydraulic model auto-calibration and continuous workflows	Surface-level treatment of cybersecurity and cloud identity	Calibration workflows are well-defined, but end-to-end data security, tenant isolation, and secure IoT device onboarding in cloud environments are rarely

							addressed together.
10	Digital Transformation in WDS based on the Digital Twins Concept (2024)	Scopus / arXiv	DT + Ensemble AI (LSTM, XGBoost)	Historical supervisory and meteorological data	Combines diverse predictive models for superior anomaly detection	Validated only in offline prototype environments	High-performing ensemble models are rarely converted into event-driven cloud pipelines that ingest and process live telemetry with low latency.
11	SWI-FEED: Smart Water IoT Framework for Evaluation of Energy and Data (2024)	IEEE / Scopus	IoT + LoRaWAN + EPANET	Large-scale simulated IoT topologies (NS-3)	Thorough evaluation of network throughput and energy consumption	Completely lacks interactive visual DT interface	IoT communications and hydraulic simulation are evaluated separately; a unified cloud middleware bridging IoT telemetry with real-time visual dashboards is needed.

12	Smart Water Security with AI and Blockchain-Enhanced Digital Twins (2025)	Scopus / arXiv	Blockchain + DT + LSTM	Rural smart water distribution networks	Strong data tamper-resistance, ledger integrity, and security	Significant latency and high implementation complexity	Decentralized security mechanisms introduce unacceptable operational latency; there is a need to simplify system security using cloud-native Identity & Access Management (IAM).
13	Digitalisation of the European Water Sector for Green Transition (2023)	MDPI (Sustainability)	Sector-wide Policy Review	Enterprise European water utilities	High-level roadmap for digital transformation across the sector	Minimal focus on modern algorithmic implementations	Policy roadmaps advocate for digitalization, but fail to specify technical, cloud-native operational architectures that small-to-medium utilities can adopt.

14	Machine Learning Applications in Sustainable Water Resource Management (2023)	Springer	Machine Learning Survey	Public hydrological & municipal datasets	Comprehensive catalog of conventional ML models in hydrology	Generic focus spanning broad environmental water resources	Existing reviews treat water resources broadly; model selection, real-time feature engineering, and cloud deployment pipelines specific to urban pipe networks require focused study.
15	Digital-Twin-Based Management of Sewer Systems: KaSyTwin Project (2025)	MDPI (Water)	Physical DT Architecture	Urban wastewater and sewer networks	Demonstrates real-world infrastructure deployment strategy	Tailored exclusively to wastewater dynamics and gravity flow	Advanced DT architectures developed for sewer networks are rarely adapted or benchmarked for pressurized, closed-conduit water distribution networks.